

LESSON 6.6

USING TWO-COLUMN TABLES OF EQUIVALENT RATIOS



LESSON INTRODUCTION

MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.

LEARNING TARGETS

- I can use two-column tables to display equivalent ratios.
- I can solve problems using two-column tables of equivalent ratios.

BENCHMARK COHERENCE

BUILDING ON

MA.5.AR.3.1 Given a numerical pattern, identify and write a rule that can describe the pattern as an expression.

WORKING TOWARD

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

ESSENTIAL QUESTIONS

- How can I use two-column tables to display equivalent ratios?
- How can I solve problems using two-column tables of equivalent ratios?
- When is a two-column table a better representation than a double number line?

LESSON NARRATIVE

Students explore equivalent ratios through the lens of two-column tables by comparing and contrasting these models to a double number line. Students should be making connections for the application of multiple representations as they transition from double number lines in previous lessons to three-column tables in the next lesson.

COMPONENT SUMMARY

6.6.1 WARM-UP (~5 MIN.)

Students consider different ways of representing division

6.6.3 GUIDED INSTRUCTION (15 MIN.)

Students use a two-column ratio table to determine the rate of pay

6.6.5 YOUR TURN! (5 MIN.)

Students use a two-column ratio table to determine the number of orbits of the International Space Station and cost of almonds

6.6.2 EXPLORATION (15 MIN.)

Students explore the advantages of modeling an equivalent ratio in a two-column table over a double number line

6.6.4 GUIDED PRACTICE (5 MIN.)

Students use a two-column ratio table to determine the rate of a runner

6.6.6 WRAP-UP (~5 MIN.)

Formative assessment opportunity using a two-column ratio table to solve real-world rate problems

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Ratio

MATERIALS

- (Optional) Note cards
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not independently recognize or understand the advantages of using a two-column table over using a double number line.
- Students may multiply instead of divide (or vice versa) when calculating equivalent ratios. For example, in 6.6.5 Your Turn! section, students may divide by $\frac{1}{3}$ instead of multiplying.

THINGS TO CONSIDER

- Explicitly show and explain the features of two-column tables that make them more flexible than double number lines:
- (1) On a double number line, the differences between numbers are represented by lengths. This can be a limitation when large or small numbers are involved. (2) A double number line dictates the ordering of the values on the line, but in a table, pairs of values can be written in any order.
- If time is an issue, then consider assigning 6.6.5 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

Consider structuring 6.6.5 Your Turn! with the

Think-Pair-Share routine. First, have students answer the questions individually. Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning. Discussing the questions will help students synthesize their understanding and build student confidence.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

The two-column tables help students identify and use the equivalent ratios to solve real-world problems. Students identify the equivalent ratios in the two-column tables. Then, they use the equivalent ratios to solve for unknown values. After solving a few problems, students will develop generalizations based on the similarities found among the problems.

DIFFERENTIATE INSTRUCTION

Students will continue engaging with rates and ratios using decimals in 6.6.5 Your Turn! and 6.6.6 Wrap-Up, as benchmarks include fluency with decimals and fractions. While a calculator can sometimes be used as scaffolding to focus on concept rather than content, students should continue building fluency as practice with decimals is spiraled throughout the unit.

6.6.1 WARM-UP: DIFFERENT WAYS OF EXPRESSING DIVISION

Component Overview: Students consider different ways division can be represented and explain connections between division and ratios.

TEACHER MOVES

Student answers will vary, and the goal isn't for all students to get this "right." The purpose is to expose students to differences in the ways people in different parts of the world express mathematics.

TEACHER MOVES

After students have written their answers, consider having students Turn and Talk or share whole-group other ways they have personally seen division represented, either from personal experience or from others.



6.6.1 Warm-Up: Different Ways of Expressing Division

In many Latin American countries and other countries around the world, the colon (:) is used instead of $\div$ to express division. Therefore, the expression "28 divided by 4" would be expressed as 28:4 instead of $28 \div 4$.

- Using what you have learned about ratios, explain how 28:4 and $28 \div 4$ can be used to express the same value.

Answers will vary.

The ratio 7:1 is equivalent to 28:4. It can be found by dividing both 28 and 4 by 4. 28 divided by 4 is 7.

- Write one question that comes to your mind after reading the information shown.

Answers will vary.

Are there any other symbols that are different in other countries that mean the same thing?

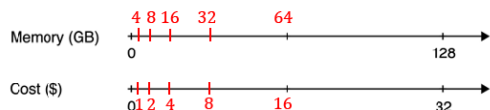


6.6.2 Exploration: Double Number Lines Versus Two-Column Tables

Work with your partner on the following.

In 2016, 128 gigabytes (GB) of portable computer memory cost \$32.

A double number line diagram that represents the situation is shown.



- One set of tick marks has already been drawn to show the result of multiplying 128 and 32 each by $\frac{1}{2}$. Label the amount of memory and the cost for these tick marks.
- Next, keep multiplying by $\frac{1}{2}$ and drawing and labeling new tick marks, until you can no longer clearly label each new tick mark with a number.
- Discuss with your partner whether or not it's possible to clearly represent the cost of 1 GB of memory on the double number line diagram.
- If it is possible, circle the cost of 1 GB of memory on the double line diagram. If it is not possible, explain why not.

Answers will vary. One would need to circle in between 0 and 1. There isn't enough space to label this clearly on the double line diagram.

- A table representing the same situation is shown. Work with your partner to find the cost of 1 GB. Use as many or few rows as needed.

Memory (gigabytes)	Cost (dollars)
128	32
64	16
32	8
16	4
4	1
1	0.25

Students may choose to use more rows to arrive at this answer. However, if they understand how to find the unit rate at this point (which many will), they may recognize that they can do it in one step by dividing both values by 128.

- Explain your steps for determining the cost per gigabyte using the table.
Answers will vary. Divide 128 by 2 and until you get to 16 and then begin dividing by 4.
- Use either the double line diagram or the ratio table to determine the cost of 512 GB of portable computer memory at this rate.

Memory (gigabytes)	Cost (Dollars)
128	32
512	128

Multiply both values by 4. The cost for 512 gigabytes is \$128.

- Discuss with your partner which representation of the equivalent ratios you preferred using to find the costs of 1 GB and 512 GB. Summarize your discussion.
I preferred to use a table because it helps me to visualize the number relationships.

6.6.2 EXPLORATION: DOUBLE NUMBER LINES VERSUS TWO-COLUMN TABLES

Component Overview: Students work with a partner to explore and compare modeling equivalent ratios in two-column tables versus double number lines. Students then have a choice of which representation to use to answer the last question, explaining their choice in a written explanation.

ENRICHMENT

Have students intentionally compare and contrast the usefulness of tools like double number lines and tables. When would a double number line be helpful? When would a two-column table be better?

- On a double number line, the differences between numbers are represented by lengths. This can be a limitation when large or small numbers are involved.
- A double number line dictates the ordering of the values on the line, but in a table, pairs of values can be written in any order.

DIFFERENTIATE INSTRUCTION

Consider allowing students to share their answers verbally prior to writing their answers. Sharing their answer verbally will allow them to rehearse their explanation and receive feedback. This will improve their confidence and the quality of their responses.

TEACHER MOVES

Poll the Class

Consider polling the class to see which students prefer the table and which prefer the double number line. This can lead to a discussion of the advantages of both representations, and the teacher can highlight when it makes the most sense to use each representation.

6.6.2 EXPLORATION: DOUBLE NUMBER LINES VERSUS TWO-COLUMN TABLES (CONTINUED)

DIFFERENTIATE INSTRUCTION

Share history or a clip from the movie *Hidden Figures* to engage more learners in the impact of IBM and important mathematical figures on our technology today.

Mathematician Mary Jackson, pictured below, is one of the key mathematical figures featured in the movie. She and women like her performed all calculations by hand in the early pre-computer days at NASA.



The IBM 7094 was regarded as one of the most powerful advanced mainframe computers in the early 1960s. NASA and the Air Force used it in the Gemini and Apollo space programs.

Despite its large size and historical impact, the IBM 7094 only had a core memory of about 150 kilobytes (KB). That's only enough to manage a few Microsoft Word documents.



Today, you can get a small USB flash drive like the one shown for less than \$20 that can store 4 gigabytes (GB) of memory.



That's almost 27,000 times the memory as the IBM 7094, and you can fit it in your pocket!



6.6.3 Guided Instruction: Using Two-Column Ratio Tables

Both double number lines and tables can be useful tools for finding equivalent ratios, but there is an important distinction between the two representations. On a double number line, the numbers on each line are listed in order.

With a table, ratios can be written in any order. For this reason, sometimes a table is easier to use to solve a problem.

Lin is paid \$90 for 5 hours of work. She used the following table to calculate how much she would be paid at this rate for 8 hours of work.

	Amount Earned (\$)	Time Worked (hours)	
$\frac{1}{5} \times$	90	5	$\times \frac{1}{5}$
	18	1	
$8 \times$	144	8	$\times 8$

- What is the meaning of the ratio that appears in the second row of the table, 18:1?
Lin is paid \$18 for 1 hour of work.
- Why was the number $\frac{1}{5}$ used as a factor?
 $\frac{1}{5}$ is used to calculate how much Lin would be paid in 1 hour. $\frac{1}{5}$ is the multiplicative inverse of 5.

- How else could the values in the second row be determined from the first row?
One could divide both columns by 5 in order to determine how much Lin is paid for each hour that she works.

- Complete the statements to explain how Lin used this table to solve the problem.

Lin used equivalent ratios to solve the problem. Lin first determined the rate for
☐ 1 hour
☐ 5 hours
☐ 8 hours
 of work by multiplying

each value in the given ratio by $\frac{1}{5}$. She then used that unit rate to find how much she would be paid for 8 hours by

multiplying the \$18 by 8.

- At this rate, how much would Lin be paid for 2.25 hours of work?
Lin would be paid $2.25 \times \$18 = \40.50 .

6.6.3 GUIDED INSTRUCTION: USING TWO-COLUMN RATIO TABLES

Component Overview: Students learn how to use a two-column ratio table to determine equivalent ratios and use them to solve problems.

DIFFERENTIATE INSTRUCTION

The table provides a visual entry point. Seeing the factors on the side of the table provides clarity for students as they try and determine the relationship between the amount of money earned and time worked in the table.

TEACHER MOVES

Have students work with a partner to answer questions 1-3 based on the table. As students discuss with their partners, circulate the room and observe any common misconceptions or understandings.

Consider employing a Think-Aloud to review student responses or guide instruction for applying concepts of ratios to tables as seen here, in 6.6.4 Guided Practice, and in the 6.6.6 Wrap-Up.

ENRICHMENT

To ensure students have a conceptual understanding of the table, consider asking the following questions:

- Why does Lin determine the rate for 1 hour of work?
- How is the rate for 1 hour helpful in determining the values for the rest of the table?
- How can you determine how much Lin will be paid for 40 hours of work?

6.6.4 GUIDED PRACTICE: USING TWO-COLUMN RATIO TABLES

Component Overview: Students use a two-column ratio table to determine the running rate of a runner and then apply this rate to determine how long it will take her to run 7.5 miles.

DIFFERENTIATE INSTRUCTION

Having students discuss the question prompts provides an auditory entry point. The two-column table and arrows with the number by which each row is multiplied provide a visual entry point.

TEACHER MOVES

Notice the similarity between this component and 6.6.6 Wrap-Up. Consider monitoring to determine what intervention might look like during 6.6.5 Your Turn!.



6.6.4 Guided Practice: Using Two-Column Ratio Tables

- Rosalie runs $\frac{1}{3}$ of a mile in 4 minutes. As part of her marathon training, her goal today is to run 7.5 miles at this rate. Use the table to determine how long it will take her to run 7.5 miles at this rate. Show how the values in each row are determined using arrows and multiplication.

Distance (miles)	Time (minutes)
$\frac{1}{3}$	4
1	12
7.5	90

Arrows and multipliers are shown to the left and right of the table:

- From $\frac{1}{3}$ to 1: multiplier 3 (indicated by a red arrow and "3 ·")
- From 4 to 12: multiplier 3 (indicated by a red arrow and "· 3")
- From 1 to 7.5: multiplier 7.5 (indicated by a red arrow and "7.5 ·")
- From 12 to 90: multiplier 7.5 (indicated by a red arrow and "· 7.5")

It will take her 90 minutes to run 7.5 miles.



6.6.5 Your Turn!

- The International Space Station (ISS) orbits around the Earth at a constant speed.



The ISS completes 2 orbits around the Earth in approximately 3 hours. Complete the table to find the approximate number of times the ISS travels around the Earth in 1 hour and in 1 day (24 hours).

Orbits around Earth Completed	2	$\times \frac{1}{3}$	$\times 24$	16
Time (hours)	3	$\times \frac{2}{3}$	$\times 1$	24

- At a local natural food store, customers can measure and purchase foods in any quantity. Patricia purchased 4 pounds of almonds for \$38.48. Greg only needs to purchase 1.5 pounds of almonds. Use the table to determine the cost for 1.5 pounds of almonds at the natural food store

Almonds (pounds)	Cost (dollars)
4	38.48
1	9.62
1.5	14.43

It will cost Greg \$14.43 to purchase 1.5 pounds of almonds.



6.6.6 Wrap-Up: Ticket Out the Door

- Julio participated in a walk-a-thon to raise money for cancer research. Walking at a constant speed, Julio travelled 6.4 miles in the first 2 hours of the walk-a-thon. His goal is to walk for 7 hours. Use the table to find how far Julio will have walked if he reaches his 7-hour goal. Show how the values in each row are determined.

Hours	Miles Walked
2	6.4
1	3.2
7	22.4

Julio will walk 22.4 miles in 7 hours.

6.6.5 YOUR TURN!

Component Overview: Students work independently to use a two-column ratio table to find additional ratios for given scenarios.

ENRICHMENT

Students may struggle to use and understand the horizontal table. Consider the following questions to address misconceptions:

- What factor can you multiply 3 hours by to find 1 hour?
- How can you use this to find the orbits completed around the earth in 1 hour?
- How is this similar to the vertical tables used in previous components?
- What factor did you multiply the given rate of 2 orbits to 3 hours by to determine the number of orbits in 24 hours?

6.6.6 WRAP-UP: TICKET OUT THE DOOR

Component Overview: Students demonstrate their understanding of using two-column ratio tables to solve real-world rate problems by determining the distance traveled when walking at a given rate.

DIFFERENTIATE INSTRUCTION

Student responses can be used for formative assessment to plan for remediation and support in the next lesson.